DUAL MOTHERBOARDS

The Dual Motherboard configuration in the Monkey Head Project combines two distinct high-performance platforms to maximize computational flexibility and redundancy. This arrangement ensures that specialized workloads are allocated to the hardware best suited for the task.

Primary Motherboard – Supermicro X9QRI-F+:

- Equipped with four Intel Xeon E5-4627 V2 processors for high-throughput parallel computing.
- Ideal for large-scale AI model training, simulation, and data processing.
- Emphasizes stability and reliability for long-duration operations.

Secondary Motherboard – ASUS ROG Zenith Extreme Alpha:

- Powered by an AMD Ryzen Threadripper 1950X with 16 cores and 32 threads.
- Optimized for multi-threaded efficiency, overclocking potential, and real-time processing.
- Supports additional GPUs or accelerators for specialized workloads.

Shared Resources:

- Hybrid memory setup with both ECC and non-ECC RAM for a balance of speed and error correction.
- Custom liquid cooling system spanning both boards to maintain thermal stability under load.
- Independent and redundant power supplies to ensure continuous operation.

Governance Integration:

- Workload assignments between boards are logged in the Unified Audit Log.
- Hardware changes and performance metrics are reviewed under governance protocols.

Testing Objectives:

- Validate cross-platform workload distribution efficiency.
- Monitor thermal and power performance under maximum load conditions.
- Assess recovery procedures in the event of a failure on one motherboard.